

Quick Activity 2: Canteen Group Order

A canteen sells **rice meals** with a fee of **₱85 each** and **drinks** with a fee of **₱25 each**. Every order receives a **10% discount** on the **rice-meal fee only**. The canteen also charges a **₱5 packaging fee per rice meal**.

Decomposition

Input	Process	Output
Number of rice meals, number of drinks, payment	Determine meal subtotal, drink subtotal, discount, packaging fee, final bill, and change	Total Fee and change

Pseudocode

START

```
READ riceMeals, drinks, payment
mealFee = riceMeals × 85
drinkFee = drinks × 25
discount = mealFee × 0.10
packagingFee = riceMeals × 5
finalFee = mealFee - discount + drinkFee + packagingFee
change = payment - finalFee
DISPLAY finalFee
DISPLAY change
```

END

Dry Run Calculations Test

Test Case 1

Input

- Rice Meals = 3
- Drinks = 4
- Payment = ₱500.00

Step	Variable	Calculation	Result
1	Meal Subtotal	3×85	₱255.00
2	Drink Subtotal	4×25	₱100.00
3	Discount	$255 \times 10\%$	₱25.50
4	Packaging Fee	3×5	₱15.00
5	Final Bill	$255 - 25.50 + 100 + 15$	₱344.50
6	Change	$500 - 344.50$	₱155.50

Output

Final Bill = ₱344.50

Change = ₱155.50

Test Case 2

Input

- Rice Meals = 5
- Drink = 3
- Payment = ₱600.00

Step	Variable	Calculation	Result
1	Meal Subtotal	5×85	₱425.00
2	Drink Subtotal	3×25	₱75.00
3	Discount	$425 \times 10\%$	₱42.50
4	Packaging Fee	5×5	₱25.00
5	Final Bill	$425 - 42.5 + 75 + 25$	₱482.50
6	Change	$600 - 482.50$	₱117.50

Output

Final Bill = ₱482.50

Change = ₱117.50

Explain: Why would applying the 10% discount to the entire bill produce an incorrect result?

If we apply the 10% discount to the entire bill it would be incorrect because the discount is meant only for rice meals, not for drinks or packaging fees. If the whole bill is discounted, it would unfairly reduce the price of items that should be charged at full value. This mistake would lower the total amount more than intended, giving the customer an unintended extra discount and producing inaccurate results in the calculation.